

Gravitational Penetration Evidence Chain: A Century of Experimental Constraints

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Abstract

When a gravitational field encounters macroscopic matter, is it absorbed, reflected, or fully penetrating? This paper systematically reviews experimental evidence from the past century, establishing a complete evidence chain: (1) laboratory torsion balances and lunar occultation analyses have excluded gravitational absorption to extremely high precision; (2) lunar laser ranging constrains any residual absorption coefficient to the 10^{-22} m²/kg level, equivalent to complete penetration; (3) precision tests of the equivalence principle (MICROSCOPE, Eöt-Wash, LLR) demonstrate that gravity penetrates matter without composition-dependent variation; (4) wide binary stars in the low-acceleration regime show indications of gravitational enhancement (Chae 2023 reports 8.7σ), but this signal remains controversial. This paper aims to provide a solid experimental foundation for future theoretical work, without presupposing any unverified theoretical framework.

1. Introduction

Gravity is the only fundamental interaction that manifests as attraction at all scales, yet what happens when a gravitational field encounters matter is rarely systematically questioned. If gravity were absorbed like light, a "gravitational shadow" should appear behind the Earth; if reflected, the equivalence principle would be violated. These questions can be answered directly by experiment.

This paper organizes evidence into three tiers:

- **Tier 1:** Gravity is not absorbed or reflected by matter (strong experimental constraint)
- **Tier 2:** Gravity fully penetrates matter without composition-dependent variation within the Solar System (strong experimental constraint)
- **Tier 3:** In low-acceleration/low-density regimes, penetration may be accompanied by modulation of the effective gravitational coupling (open question, pending test)

2. Tier 1: Gravity Is Not Absorbed or Reflected

2.1 Theoretical Expectations and Experimental Tests

Two distinct classes of gravitational shielding models have appeared in history. Le Sage (1784) proposed a corpuscular impact model, in which gravity arises from isotropic particle flux pushing objects, with matter 遮挡 producing a "gravitational shadow." Majorana (1919) proposed a different attenuation model, assuming gravitational flux decays exponentially through matter, quantified by an absorption coefficient h .

These models differ mechanistically but share the common prediction that gravity can be attenuated by matter, allowing a unified constraint via h .

2.2 Laboratory Torsion Balance Experiments

Majorana himself measured $h \leq 4.3 \times 10^{-14} \text{ cm}^2/\text{g}$ (equivalent to $4.3 \times 10^{-15} \text{ m}^2/\text{kg}$) using lead balls and a torsion balance. Subsequent improvements (Kreuzer 1966, Panov 1985) pushed the limit to $h \leq 10^{-15} \text{ m}^2/\text{kg}$. Savrov (2012) systematically confirmed the validity of these data and unified the definition standards across different 文献.

2.3 Lunar Occultation Gravity Anomaly Analysis

During lunar eclipses, if the Moon shields solar gravity, ground gravimeters should register anomalies. Multiple analyses in the 1970s–1990s (Saxl & Allen 1971, Van Flandern 1975) found no such anomalies, pushing the h upper limit down to $6 \times 10^{-19} \text{ m}^2/\text{kg}$.

2.4 Lunar Laser Ranging (LLR): The Strongest Constraint

Fifty years of continuous LLR data track the Moon's orbit to test whether the Earth and Moon fall toward the Sun with equal acceleration. Williams, Turyshev & Boggs (2009) obtained:

$$h = (3 \pm 5) \times 10^{-22} \text{ m}^2/\text{kg}$$

This is the strongest constraint to date, 7 orders of magnitude tighter than laboratory results and 3 orders tighter than occultation analyses.

Conclusion: Gravitational shielding (absorption/reflection) is excluded to extremely high precision. Gravity must fully penetrate

matter.

3. Tier 2: Complete Penetration Without Composition Dependence

3.1 Weak Equivalence Principle: MICROSCOPE

The French MICROSCOPE satellite (2016–2018) placed titanium and platinum test masses in a 710 km orbit, measuring their differential free-fall acceleration. Touboul et al. (2022) reported:

$$\eta = (a_{\text{Ti}} - a_{\text{Pt}}) / (a_{\text{Ti}} + a_{\text{Pt}}) \leq 2.7 \times 10^{-15}$$

Gravity acts identically on different compositions to better than 1 part in 10^{15} .

3.2 Eöt-Wash Torsion Balance

The Eöt-Wash series (Schlamminger et al. 2008) used a rotating torsion balance to compare accelerations of beryllium and titanium, giving $\eta \leq 10^{-13}$ across multiple material combinations.

3.3 Strong Equivalence Principle: LLR

The SEP requires that gravitational self-energy also participates normally in free fall. The Earth's gravitational self-energy is about 4.6×10^{-10} of its mass. LLR constrains any differential acceleration to:

$$(M_G / M_I - 1) = (2 \pm 5) \times 10^{-13}$$

Conclusion: At Solar System scales, gravity fully penetrates matter and acts identically on all compositions. The equivalence principle holds to 10^{-15} .

4. Tier 3: Indications of Modulation in the Low-Acceleration Regime

4.1 Wide Binary Anomaly

Wide binaries are stellar pairs separated by thousands of AU, orbiting with accelerations below 10^{-9} m/s²—ideal laboratories for testing gravity in the low-acceleration regime.

Chae (2023, ApJ 952, 128) analyzed 26,615 wide binaries from Gaia DR3 and found:

$$\gamma \equiv G_{\text{eff}} / G_{\text{N}} = 1.43 \pm 0.06$$

Deviation from Newtonian gravity: **8.7σ** .

4.2 Controversy and Technical Background

Banik et al. (2024, MNRAS 527, 1148) obtained results consistent with Newtonian gravity ($\gamma = 1.12^{+0.27}_{-0.22}$, 0.4σ) using different sample selection and orbit modeling. Key 争议 points: unresolved triple systems, prior distributions, and stellar mass estimates.

Additionally, Gaia DR3 astrometric data exhibits significant excess noise from unresolved companions or instrumental systematics. This noise

effect on wide binary orbit fitting remains under investigation and is a key technical background to the controversy.

Hernandez & Kroupa (2024, arXiv:2410.17178) noted that analyses favoring Newtonian gravity have issues in sample selection and statistical methodology, while the anomaly-supporting team used more conservative samples.

4.3 Current Status

The wide binary anomaly is **not settled** but remains a significant observational indication. It may arise from systematics, unknown tertiary contamination, or genuine new physics. If the latter, it would suggest that in extremely low-density/low-acceleration environments, gravitational penetration may be accompanied by modulation of the effective coupling strength.

Conclusion: Tier 3 evidence remains open, awaiting independent data. Gaia DR4 (expected late 2026 to early 2027) will provide $\sim 10^6$ astrometric orbit solutions, potentially resolving the signal at decisive significance.

5. Evidence Chain Summary

Tier	Proposition	Status	Key Experiment
1	No absorption/reflection	Strong constraint	LLR: $h = (3\pm 5)\times 10^{-22} \text{ m}^2/\text{kg}$

Tier	Proposition	Status	Key Experiment
2	Penetration without composition dependence	Strong constraint	MICROSCOPE: $\eta \leq 2.7 \times 10^{-15}$
3	Possible modulation in low-acceleration regime	Controversial	Wide binaries: $\gamma \approx 1.43, 8.7\sigma$

6. Conclusions

1. **Gravity is not absorbed or reflected** by matter — LLR constrains any residual absorption to 10^{-22} m²/kg.
 2. **Gravity fully penetrates** matter without composition-dependent variation — the equivalence principle holds to 10^{-15} .
 3. **In low-acceleration/low-density regimes, gravitational modulation remains an open question** — the wide binary signal (8.7σ) is controversial and awaits Gaia DR4.

Note: "Penetration" is used phenomenologically—gravity shows no detectable attenuation from macroscopic matter. In General Relativity, gravity is spacetime curvature, not particle flux through matter.

References

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Author Statement: This is an experimental evidence review, presupposing no unverified theoretical framework. All conclusions are based on published experimental data.

Verification, discussion, and criticism are welcome.